



BIT1123/BISE2093/DIT1113

BIT1123 OBJECT ORIENTED PROGRAMMING

ASSIGNMENT 1 - INDIVIDUAL

(20%)

ALL SECTIONS

FACULTY OF INFORMATION TECHNOLOGY

CITY UNIVERSITY MALAYSIA

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1. Introduction

This report illustrates what I learned during my time in the Object Oriented Programming (OOP) tutorials with Java. The hands-on exercises gave me a good understanding of how to structure Java programs with classes and objects. Other course concepts introduced in the tutorials were constructors, methods, inheritance, abstract classes, and GUI programming. I'm using GitHub to store all my tutorial work in one repository and to help build better habits with handling programming projects.

2. Knowledge Gained from the Tutorials

Tutorial 1

I've learnt the fundamental concepts of Java classes and objects. Implements attributes, constructors and methods. The student example helped me understand how can we have data and behaviour inside of a class.

Tutorial 2

Practiced more classes and objects in Java. This helped me to get more comfortable with creating objects and sending values to them.

Tutorial 3–4

There are exercises on inheritance and relationships between classes in the repository. I found out that classes could share common characteristics but have distinct behaviour.

Tutorial 5

Experienced in OOP programming and documentation. This let me realize that it was important to write structured code and include supporting documentation.

Tutorial 6

Engaged in employee and lecturer classes. This was a great opportunity to get more practice using inheritance and relationships between classes.

Tutorial 7

practiced abstract classes and inheritance using appliance-related classes such as Microwave and Washing Machine. This improved my understanding of parent and specialized child classes.

Tutorial 8–9

The repository contains the work for this tutorial period. These exercises helped me continue practicing Java programming and maintain the project structure used for the course.

Tutorial 10

worked with a Java GUI quiz application using Questions.java and QuizBattleGUI.java. This introduced me to building a more interactive Java program with a graphical interface.

3. Challenges Encountered

The major difficulty was in grasping working together of several java classes. Initially, it was a bit tricky to understand the concept of a class vs an object and how constructors are used to setup the data of an object. Another challenge was inheritance; I had to learn how a parent class and child classes are related. The GUI tutorial was also difficult not only because of the size of the program structure, but because this tutorial had an even larger program structure than the previous ones.

4. How I Overcame the Challenges

I solved these difficulties by following the example tutorials and testing in between steps, to make sure it was working and behaving as expected. If there were errors then I looked at the classes names, methods, constructors and file relationships. I also created separate tutorial folders for the exercises to be able to locate and review each exercise more easily. I was able to use GitHub to manage my work in a single location and see how far I've come over the course of my projects.

A. Let's see how much my Java programming skills have improved.5. Let's see how many improvements I made to my Java programming skills.

The tutorials helped me to develop my java skills. I feel more comfortable writing the classes, declaring attributes, writing constructors, writing methods and writing objects. I also got better at reading Java code and how different files relate to each other. The later exercises took me from basic examples of classes to inheritance and abstract classes to GUI programming.

A basic grasp of the concepts of Object Oriented Programming.

I came to the understanding of OOP from the tutorials that it is a program that is organized around objects holding data and behaviour. I understood that classes are structures for objects, constructors initialize data associated with objects and methods define behaviour of an object. Also got to know the concept of inheritance, that is a child class can be based on a parent class, and abstract classes, which can give a child

class a common structure. These programming concepts have been introduced to me through the GUI quiz application which has given me an insight on how these can be applied in a more practical application.

7. Future Learning Plans

In the future, I hope to improve my programming skills in Java through the practice of larger OOP projects and that I will write more efficient, and better organized code. Also, I wish to get more confident with inheritance, polymorphism, interfaces, exception handling, file handling and GUI development. Another objective is to enhance my Git and GitHub abilities to be able to manage Programming projects in a far more professional manner.

8. Conclusion

Overall, the OOP tutorial provided me a helpful introduction on programming using java and OOP. I learned practical experience in classes, objects, constructors, methods, inheritance, abstract classes, and GUI programming. I also learned that it is important to structure code and provide documentation of my work through the use of GitHub. For some concepts I had to overcome difficulties in learning but through the continual practice I have gained an understanding and confidence in my practice. I will keep on practising java and OOP concepts to boost my programming skills.

9. GitHub Repository URL

<https://github.com/dakeenomar-design/oop>

Note: The assignment brief also asks for class code, program, and NRIC/passport number in the submission information. These details were not available in the provided repository/report information, so they should be added by the student if required.